

TDD

RED  GREEN  REFACTOR

```
===== FAILURES =====
test_endpoint_recarga_valida

def test_endpoint_recarga_valida():
    respuesta = client.get("/calcular-recarga?monto=3000&plan=premium")
    assert respuesta.status_code == 200
    assert respuesta.json()["bono_aplicado_porcentaje"] == 30
    assert 0 == 30
E
tests/unit/test_api.py:10: AssertionError

test_endpoint_recarga_invalida

def test_endpoint_recarga_invalida():
    respuesta = client.get("/calcular-recarga?monto=500")
    assert respuesta.status_code == 400
    assert 200 == 400
E
+ where 200 = <Response [200 OK]>.status_code
E
tests/unit/test_api.py:14: AssertionError

test_monto_inferior_rechazado

def test_monto_inferior_rechazado():
    # Arrange & Act & Assert
    with pytest.raises(MontoInvalidoError):
        # Act
E
Failed: DID NOT RAISE <class 'src.recargas.MontoInvalidoError'>
E
tests/unit/test_recargas.py:7: Failed

test_monto_superior_rechazado

def test_monto_superior_rechazado():
    with pytest.raises(MontoInvalidoError):
        # Act
E
Failed: DID NOT RAISE <class 'src.recargas.MontoInvalidoError'>
E
tests/unit/test_recargas.py:12: Failed

test_calcular_bonificaciones[10000-estandar-10]

monto = 10000, tipo_plan = 'estandar', bono_esperado = 10
```

```
test_calcular_bonificaciones[10000-estandar-10]

monto = 10000, tipo_plan = 'estandar', bono_esperado = 10

@pytest.mark.parametrize("monto, tipo_plan, bono_esperado", [
    (1000, "estandar", 0),
    (9999, "estandar", 0),
    (10000, "estandar", 10),
    (10000, "premium", 15),
    (29999, "estandar", 10),
    (30000, "estandar", 25),
    (30000, "premium", 30),
    (50000, "estandar", 25),
    (1000, "premium", 5),
])
def test_calcular_bonificaciones(monto, tipo_plan, bono_esperado):
    # Act
    porcentaje_obtenido = procesar_recarga(monto, tipo_plan)

    # Assert
    assert porcentaje_obtenido == bono_esperado
E
assert 0 == 10
E
tests/unit/test_recargas.py:32: AssertionError

test_calcular_bonificaciones[10000-premium-15]
```

```
tests/unit/test_recargas.py:32: AssertionError

===== warnings summary =====
venv\Lib\site-packages\fastapi\testclient.py:1
C:\Users\asus\OneDrive\Documents\Parcial-2\venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient.TestClient' is deprecated. Import 'httpx' from 'starlette.testclient' instead.
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html

===== short test summary info =====
FAILED tests/unit/test_api.py::test_endpoint_recarga_valida - assert 0 == 30
FAILED tests/unit/test_api.py::test_endpoint_recarga_invalida - assert 200 == 400
FAILED tests/unit/test_recargas.py::test_monto_inferior_rechazado - Failed: DID NOT RAISE <class 'src.recargas.MontoInvalidoError'>
FAILED tests/unit/test_recargas.py::test_monto_superior_rechazado - Failed: DID NOT RAISE <class 'src.recargas.MontoInvalidoError'>
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[10000-estandar-10] - assert 0 == 10
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[10000-premium-15] - assert 0 == 15
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[29999-estandar-10] - assert 0 == 10
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[30000-estandar-25] - assert 0 == 25
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[30000-premium-30] - assert 0 == 30
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[50000-estandar-25] - assert 0 == 25
FAILED tests/unit/test_recargas.py::test_calcular_bonificaciones[1000-premium-5] - assert 0 == 5
===== 11 failed, 2 passed, 1 warning in 1.03s =====
(parcial-2) PS C:\Users\asus\OneDrive\Documents\Parcial-2>
```

```
Terminal Local (2) Local (4) Local (3)
Windows PowerShell
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Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

(parcial-2) PS C:\Users\asus\OneDrive\Documentos\Parcial-2> uv run pytest tests/unit/
===== test session starts =====
platform win32 -- Python 3.14.0, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\asus\OneDrive\Documentos\Parcial-2
configfile: pyproject.toml
plugins: anyio-4.13.0, locust-2.44.1, bdd-8.1.0, cov-7.1.0
collected 13 items

tests/unit/test_api.py .
tests/unit/test_recargas.py .....

===== warnings summary =====
.venv\Lib\site-packages\fastapi\testclient.py:1
C:\Users\asus\OneDrive\Documentos\Parcial-2\.venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 13 passed, 1 warning in 0.60s =====
(parcial-2) PS C:\Users\asus\OneDrive\Documentos\Parcial-2>
```

LOCUS (Prueba de 30 usuarios) terminal y forma más gráfica desde el navegador:

```
(parcial-2) PS C:\Users\asus\OneDrive\Documentos\Parcial-2> uv run locust -f tests/performance/Locustfile.py --headless -u 30 -r 10 --run-time 15s --host http://127.0.0.1:8000
[2026-06-05 18:12:47.031] CUERVO-ASUS/INFO/Locust.main: Starting Locust 2.44.1
[2026-06-05 18:12:47.033] CUERVO-ASUS/INFO/Locust.main: Run time limit set to 15 seconds
Type Name # reqs # fails Avg Min Max Med req/s failures/s
-----
Aggregated 0 0(0.00%) 0 0 0 0 0.00 0.00

[2026-06-05 18:12:47.035] CUERVO-ASUS/INFO/Locust.runners: Ramping to 30 users at a rate of 10.00 per second
Type Name # reqs # fails Avg Min Max Med req/s failures/s
-----
GET /calcular-recarga?monto=10045&plan=estandar 1 0(0.00%) 50 50 50 50 0.00 0.00
GET /calcular-recarga?monto=11129&plan=premium 1 0(0.00%) 10 10 10 10 0.00 0.00
GET /calcular-recarga?monto=13073&plan=premium 1 0(0.00%) 52 52 52 52 0.00 0.00
GET /calcular-recarga?monto=14006&plan=premium 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=15108&plan=estandar 1 0(0.00%) 12 12 12 12 0.00 0.00
GET /calcular-recarga?monto=16027&plan=estandar 1 0(0.00%) 2 2 2 2 0.00 0.00
GET /calcular-recarga?monto=16696&plan=premium 1 0(0.00%) 85 85 85 85 0.00 0.00
GET /calcular-recarga?monto=19099&plan=premium 1 0(0.00%) 2 2 2 2 0.00 0.00
GET /calcular-recarga?monto=20640&plan=premium 1 0(0.00%) 1 1 1 1 0.00 0.00
GET /calcular-recarga?monto=21610&plan=estandar 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=22193&plan=estandar 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=23030&plan=premium 1 0(0.00%) 10 10 10 10 0.00 0.00
GET /calcular-recarga?monto=25075&plan=estandar 1 0(0.00%) 3 3 3 3 0.00 0.00
GET /calcular-recarga?monto=27738&plan=estandar 1 0(0.00%) 54 54 54 54 0.00 0.00
GET /calcular-recarga?monto=31936&plan=premium 1 0(0.00%) 2 2 2 2 0.00 0.00
GET /calcular-recarga?monto=35019&plan=premium 1 0(0.00%) 52 52 52 52 0.00 0.00
GET /calcular-recarga?monto=38708&plan=premium 1 0(0.00%) 1 1 1 1 0.00 0.00
GET /calcular-recarga?monto=41655&plan=estandar 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=43093&plan=premium 1 0(0.00%) 2 2 2 2 0.00 0.00
GET /calcular-recarga?monto=44786&plan=estandar 1 0(0.00%) 54 54 54 54 0.00 0.00
GET /calcular-recarga?monto=45918&plan=premium 1 0(0.00%) 50 50 50 50 0.00 0.00
GET /calcular-recarga?monto=46346&plan=premium 1 0(0.00%) 11 11 11 11 0.00 0.00
GET /calcular-recarga?monto=46800&plan=premium 1 0(0.00%) 1 1 1 1 0.00 0.00
GET /calcular-recarga?monto=47657&plan=estandar 1 0(0.00%) 50 50 50 50 0.00 0.00
GET /calcular-recarga?monto=54935&plan=estandar 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=55986&plan=premium 1 0(0.00%) 53 53 53 53 0.00 0.00
GET /calcular-recarga?monto=57996&plan=premium 1 0(0.00%) 9 9 9 9 0.00 0.00
GET /calcular-recarga?monto=59298&plan=estandar 1 0(0.00%) 1 1 1 1 0.00 0.00
GET /calcular-recarga?monto=94756&plan=premium 1 0(0.00%) 52 52 52 52 0.00 0.00
Aggregated 29 0(0.00%) 23 1 85 10 0.00 0.00
```

Barra de marcador... HTML y CSS OM IA U Otros Yo Hostar enmobiliaria

LOCUST

Hosthttp://127.0.0.1:8000

StatusRUNNING

Users30

RPS20.02

Failures0%

EDIT

STOP

RESET

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/calcular-recarga?monto=10710&plan=estandar	1	0	2.48	2	2	2.48	2	2	147	0.17	0
GET	/calcular-recarga?monto=10778&plan=premium	1	0	16.58	17	17	16.58	17	17	146	0.17	0
GET	/calcular-recarga?monto=10792&plan=premium	1	0	2.47	2	2	2.47	2	2	146	0.17	0
GET	/calcular-recarga?monto=11038&plan=estandar	1	0	2.72	3	3	2.72	3	3	147	0.17	0
GET	/calcular-recarga?monto=11604&plan=estandar	1	0	3.58	4	4	3.58	4	4	147	0	0
GET	/calcular-recarga?monto=11637&plan=estandar	1	0	2.47	2	2	2.47	2	2	147	0.17	0